MIL-R-37053B
17 May 1978
SUPERSEDING
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7 June 1974

MILITARY SPECIFICATION

REFRACTOMETER, Hand, Aluminum

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a lightweight hand refractometer used to measure specific gravity of human urine and serum protein concentration (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, CODE-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6650

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Aluminum. Aluminum shall be 2011 aluminum alloy with a clear anodized finish.

3.1.2 Glass. Glass shall be clear, colorless and of optical quality.

3.2 Design and construction. Refractometer shall be a lightweight, tubular-shaped monocular optical instrument, fabricated of aluminum, designed to measure refractive index, and shall be calibrated to provide scales for direct reading of specific gravity of human urine, serum protein concentration, and refraction (see 6.3). Refractometer shall have an optical system consisting of a measuring prism, an objective, a reticle, and a focusable eyepeice, and shall have a plastic cover plate. Refractometer shall be a transmission type, designed to provide maximum contrast between light and dark. Overall dimensions shall be 6-3/8 ± 1/4 inches (in.) in length and 1-1/4 ± 1/8 in. in diameter. Instrument shall be provided with means for recalibration, and shall be contained in a cushioned protective case. Weight of the unit shall be not more than 11 ounces.

3.2.1 Optical system.

3.2.1.1 Measuring prism. Measuring prism shall consist of a hermetically sealed glass prism which contains a clear homogenous stable liquid and an air bubble. Shall also include a prism cover (sealed glass plate). Index of refraction of the liquid shall change with temperature in a manner which provides automatic compensation for temperature variations. Air bubble shall be of sufficient size to allow for expansion and contraction of liquid due to temperature variation. When tested as specified in 4.4.1 refraction scale shall read or shall be capable of being calibrated to read the same as the distilled water or the standard.

3.2.1.1.1 Prism cover. Prism cover shall be a framed clear optical quality glass plate, forming a window at the distal end of the instrument. Overall dimensions shall be 1-9/16 ± 1/16 in. long, 5/8 ± 1/32 in. wide and 1/8 ± 1/32 in. thick. Glass plate shall have flat, optically polished, parallel surfaces consistent with the intended use and accuracy of the instrument. Glass plate shall be a specimen sample holder and window for the beam of light entrained through the optical system.

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3.2.1.2 Objective. The objective glass lens, a component of the optical system in the refractometer body tube, shall be designed so as to direct a beam of light in a true path to the scales on the reticle (see 3.2.1.3).

3.2.1.3 Reticle. Reticle shall consist of a magnifying glass lens containing three direct reading scales, one for urine specific gravity, one for serum protein concentration in grams/100 milliliters, and one for refraction. With sample specimen in place, the optical system shall produce a dark area on the reticle. The highest reading on the visible portion (bright area) of the applicable scale on the reticle shall be the quantitative value of the sample specimen.

3.2.1.3.1 Scales. Scales shall be effected on the reticle lens by a photo etch process with a chrome deposit. Scales shall be clear and distinctly readable when viewed through the focusable eyepiece (see 3.2.1.4). Specific gravity scale shall read from 1.000 to 1.035 and shall be graduated in 0.001 intervals, with values marked at each 0.005 graduation. Serum protein scale shall be from 2.5 to 15.0 grams/100 milliliters and shall be graduated in 0.1 grams/100 milliliter intervals with values marked at each 0.5 graduation. Refraction scale shall be from 0 to 320, shall be graduated in increments of 2 and every fifth increment shall be numbered. Scales shall be coordinated as shown in table I.

TABLE I. Scale coordination.

Refraction	Urine Specific Gravity
0	= 1.000
60	= 1.017
	Serum or protein plasma
96	> 3.9 < 4.0
126	> 5.5 < 5.6
242	= 11.0

3.2.1.4 Eyepeice. Eyepeice at the proximal end of the instrument shall be a lens having a magnification of 10X. The lens element shall be made of clear optical glass and shall be low-reflection coated. Eyepeice shall be focusable to compensate for differences in eyesight characteristics and shall magnify the scales on the reticle for ease of readability.

3.3 Coverplate. Coverplate of clear methyl methacrylate molded resin shall be of such size as to adequately cover the glass plate (see 3.2.1.1.1) and shall have an overall thickness of 7/32 ± 1/16 in. Semicircular indentations centrally located on the upper and lower ends of the coverplate shall be provided to facilitate placing sample specimen on the exposed portion of the glass plate so that the liquid is drawn between the two facing surfaces. Exposed semicircular glass plate surface, at lower indentation of coverplate, also shall afford admittance of light to the optical system. Coverplate shall be mounted on corrosion-resisting metal wire supports so as to be capable of readily being swung onto, or completely away from, the glass plate. In position of use, coverplate and glass plate facing surfaces shall accommodate a thin layer of liquid sample. With coverplate swung away from the glass plate, the refractometer shall also be capable of being used by immersion into sample specimen.

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3.4 Calibration. Instrument shall be provided with means for in-field calibration in the form of a setscrew beneath the objective lens which shall be readily set with a suitable screwdriver. The opening over the setscrew shall be filled with a suitable sealer to prevent leaking of moisture into the instrument.

3.5 Case. Case for refractometer shall be sheath-type and shall be made of 40-point, black vinyl flock material. Case shall be lined with a 1/4-in. nominal thickness, 2-pound density, dark-colored urethane foam to afford adequate protective cushioning for the instrument. Case shall have a flap-type cover with adequate means for effecting a secure closure. Centrally positioned on the proximal end of the back side of the case shall be a nominal 2-1/4-in., sturdy, pocket-type clip.

3.6 Marking.

3.6.1 Refractometer. Refractometer shall be permanently and legibly marked with the following information:

National Stock Number
Manufacturer's name or registered trademark
Model number
"U.S."

3.6.2 Case. Case shall be permanently marked with the following information:

Item Identification
Manufacturer's name or registered trademark

3.7 Service data. The contractor shall furnish, with each unit, two copies (or sets) of a technical manual which contains complete instructions for installation operation and maintenance.

3.8 Workmanship. Workmanship shall be first class throughout. Instrument and component parts shall be finished smoothly and shall be free from defects listed in table IV and other defects which detract from their appearance or may impair serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the aluminum and glass may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. For examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-2	2.5

4.2.2 For tests. All refractometers shall be tested in accordance with table III. Refractometers which fail accuracy test shall be rejected. Tests shall be performed on end item or during production.

TABLE III. For tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Refractometer	Accuracy	3.2.1.1	4.4.1	100%

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4.3 Examination. The refractometer shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with, but shall not be limited to, the defects listed in table IV.

TABLE IV. Classification of defects.

Category	Defects
Major	
101	Component missing.
102	Glass parts not free of cracks, chips or breaks.
103	Plastic parts not free of cracks, chips or breaks.
104	Metal parts not free of corrosion.
105	Component parts not free of sharp edges.
106	Hole above calibration screw not sealed.
107	Plastic cover plate not free of deformation.
108	Flap case (cover) not capable of being securely closed.
109	Case not free of cracks, rips, tears or sharp edges.
110	Printing of scales illegible or incomplete.
111	Glass parts not free of scratches, striae, seeds or marks.
Minor	
201	Marking on instrument and case illegible, incomplete or missing.
202	Instrument surfaces not smoothly finished.
203	Lining (foam plastic) not adhering to the case.
204	Clip not securely attached to back of case.
205	Case not free of rough edges.

4.3.2 Dimensional examination. The refractometer shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Each unit shall be tested to assure that it is in operational condition before shipment.

4.4.1 Accuracy. Refraction scale shall be tested at zero and at a value above 200. Testing shall be performed by comparison with distilled water and other standards. Refractometer shall be operated in accordance with instructions in service data. Refraction standards shall be verified with a master refractometer at 20^{0} C.

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Refractometers shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Refractometers shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Each refractometer shall be positioned within the case and packaged in accordance with MIL-P-116, method IC-2.

5.1.2 Level C. Refractometers shall be clean and dry. Each refractometer shall be positioned with the case and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Twenty-four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twenty-four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged refractometers shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These containers shall conform to the carrier's rules and regulations.

5.3 Marking. Unit packages and exterior or shipping containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 The refractometer covered by this specification is listed in the Federal Supply Catalog as follows:

National Stock Number	Item Identification
6650-00-933-3218	REFRACTOMETER, Hand, Aluminum

6.3 Refraction scale. The refraction scale varies directly with the difference in refractive index as illustrated in the following table.

Refractive index	Refraction
1.3330	0
1.3340	10
1.3650	320

Distilled water which has refractive index of 1.3330 at 20°C is zero on the refraction scale.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6650-0095

Review activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-R-37053B REFRACTOMETER, Hand, Aluminum

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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